

FEDERAL STATISTICAL SYSTEM · AI READINESS

The FSS AI Vocabulary

Authoritative, plain-language definitions of AI terms for the Federal Statistical System — for experts in federal statistics who are new to AI.

43 terms across four tiers · generated 2026-06-22 · every definition from a public, free-access source · grounded against the fss-policy-knowledge graph.

How to read this

Each term has two layers. **Layer 1 — the official definition** tells you what the term means in the world, with its source shown. **Layer 2 — the FSS context** tells you what it means *inside* the federal statistical system. The two are always kept distinct, and every claim is labeled so you know what kind of claim it is.

Legend

- ★ **FLAGSHIP** — the project's headline term; its definition (grounded) and forward interpretation (ahead-of-policy) are shown separately.
- ⚠ **FALSE FRIEND** — means something different in AI than in FSS practice. Read these.
- ⚖ **CONTESTED** — politically live policy context.
- ✓ **GROUNDED IN THE FSS CORPUS** — backed by cited corpus documents.
- 📌 **PRACTITIONER NOTE** — a best-read interpretation, NOT corpus authority.
- ▶ **FORWARD INTERPRETATION** — dated, ahead of the documented standard; bounded from the definition.

Sourcing

Public, freely-accessible authority only. No paywalled sources (e.g. ISO/IEC standards behind a paywall are NOT used). Priority: US federal (NIST/OMB/statute) > open standards > open foreign-government or open industry/academic references > needs_source. A free, open definition always beats a gatekept one. source_quality is tagged honestly: where no US-federal definition exists, the record says so rather than dressing a foreign/industry source as federal.

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confabulation 
information integrity
automated decision system

FSS Bridge

disclosure avoidance 
statistical disclosure limitation
measurement error
nonsampling error
data asset 
confidentiality 
statistical product 
noise injection 

AI-ready data ★

High Translation-Load

hallucination 
bias 
validity 
reliability 
model card
provenance
human-in-the-loop
fitness for use
prompt

AI Core

14 terms

artificial intelligence AI CORE

also: *AI*

An umbrella term for software that does tasks we used to think needed a human — predicting, classifying, recommending, generating text — by finding patterns in data rather than following hand-written rules. There is no single legal definition; the statutory one (15 U.S.C. § 9401) is the federal anchor.

DEFINITION

A machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.

— NIST SP 800-218A (per 15 U.S.C. § 9401(3)) U.S. federal

Also defined: Any artificial system that performs tasks under varying and unpredictable circumstances without significant human oversight, or that can learn from experience and improve performance when exposed to data sets; ... designed to think or act like a human ... or that is designed to act rationally. — Sec. 238(g), John S. McCain NDAA for FY2019 (10 U.S.C. § 2358 note) U.S. federal

✓ GROUNDED IN THE FSS CORPUS

Covers any automated analytical system a statistical agency runs — imputation models, record linkage, automated coding, disclosure-avoidance engines. The label matters because OMB AI guidance (M-25-21) attaches obligations to 'AI' use.

Cited in: [eo_13859](#) , [eo_14110](#) , [m_24_10](#)

AI system AI CORE

also: artificial intelligence system

The whole working setup — the data pipeline, the model, and the output logic — not just the model file. When you evaluate AI risk, you evaluate the system, because harms come from how the pieces fit together, not the model alone.

DEFINITION

An engineered or machine-based system that can, for a given set of objectives, generate outputs such as predictions, recommendations, or decisions influencing real or virtual environments. AI systems are designed to operate with varying levels of autonomy.

— NIST AI 100-1 (AI Risk Management Framework 1.0) U.S. federal

✓ GROUNDED IN THE FSS CORPUS

Distinguishes the full deployed tool (e.g., a record-linkage system: input data + model + match-decision logic + output) from the model component. OMB risk-management practices apply to the system in use.

Cited in: [eo_14110](#) , [nist_ai_rmf_100_1](#) , [m_24_18](#)

AI model AI CORE

also: model

The trained mathematical object inside an AI system — the thing that turns inputs into outputs. Think of it like a fitted statistical model, but usually far larger and learned from data rather than specified by a methodologist.

DEFINITION

A component of an AI system that implements AI technology and uses computational, statistical, or machine-learning techniques to produce outputs from a given set of inputs.

— OMB M-24-10 / NTIA AI Accountability Policy Report glossary U.S. federal

CROSSWALK → **fitted statistical model / estimator** (analog_of): Both map inputs to outputs from data. Disanalogy: a statistical estimator has an explicit, documented functional form and known sampling properties; an AI model's form is learned and often opaque, so its error properties are empirical, not derived.

✓ GROUNDED IN THE FSS CORPUS

In the FSS, 'AI models' include imputation models, disclosure-avoidance noise mechanisms, and predictive coding tools used in survey processing.

Cited in: [eo_14319](#) , [m_24_10](#) , [m_24_18](#)

machine learning AI CORE

also: *ML*

A way of building models by letting an algorithm find patterns in data and improve with more examples, instead of a person writing the rules. Most of what people call 'AI' today is machine learning.

DEFINITION

The development and use of computer systems that adapt and learn from data with the goal of improving accuracy.

— NIST SP 800-55v1 U.S. federal

✓ GROUNDED IN THE FSS CORPUS

Increasingly used in federal surveys for donor selection in imputation, nonresponse adjustment, and automated industry/occupation coding.

Cited in: [m_24_18](#) , [eo_14110](#) , [m_24_10](#)

generative AI AI CORE

also: *GenAI, generative artificial intelligence*

AI that produces new content — text, images, code, synthetic data — that looks like its training data but is newly generated, not retrieved from a database. Large language models are the most familiar kind.

DEFINITION

The class of AI models that emulate the structure and characteristics of input data in order to generate derived synthetic content such as images, videos, audio, text, and other digital content.

— EO 14110 §3(p) U.S. federal

✓ GROUNDED IN THE FSS CORPUS

Introduces confabulation risk when drafting statistical narratives or metadata, and is an emerging tool for synthetic-data generation for privacy-protective dissemination.

Cited in: [eo_14319](#) , [m_24_10](#) , [m_24_18](#)

large language model AI CORE

also: LLM

A generative-AI model trained on very large amounts of text that predicts likely next words, letting it answer questions, summarize, translate, and draft. It is a fluent pattern-matcher over language, not a database of verified facts — which is why it can state wrong things confidently (see confabulation).

DEFINITION

a class of language models that use deep-learning algorithms and are trained on extremely large textual datasets that can be multiple terabytes in size. LLMs can be classed into two types: generative or discriminatory.

— AI Assurance (2022), via NIST AI 100-3 (AIRC glossary) academic

✓ GROUNDED IN THE FSS CORPUS

The tool most FSS staff will actually touch. Useful for drafting and summarizing; unsafe for producing statistics or facts without verification. NIST AI 600-1 (Generative AI Profile) is the governing risk reference — pull its exact definition before formal use.

Cited in: [eo_14319](#) , [eo_14110](#) , [americas_ai_action_plan_2025](#)

foundation model AI CORE

also: frontier model

A very large model pre-trained on broad data that serves as a reusable base others adapt (fine-tune) for specific tasks. Most commercial LLMs are foundation models.

DEFINITION

In generative AI, models trained on broad data using self-supervised learning that can be adapted such as through fine-tuning for a variety of downstream tasks.

— NIST AI 100-2e2025 U.S. federal

✓ GROUNDED IN THE FSS CORPUS

When an agency adopts a vendor foundation model, it inherits that model's training data, biases, and limitations — a value-chain risk that can collide with CIPSEA or SPD No. 15 category definitions.

Cited in: [m_24_10](#) , [eo_14110](#) , [m_24_18](#)

training data AI CORE

The examples a model learns from. The model's behavior, blind spots, and biases are inherited from this data — if a group is underrepresented in training data, the model tends to perform worse for that group.

DEFINITION

A dataset from which a model is learned.

— AI Fairness 360, via NIST AI 100-3 (AIRC glossary) academic

CROSSWALK → **sampling frame / sample coverage** (analog_of): Both determine who is represented in what the system 'knows.' Disanalogy: a frame's coverage is measurable against a known target population; training-data coverage often has no enumerable target, so its gaps are harder to quantify.

✓ GROUNDED IN THE FSS CORPUS

Directly analogous to a sample frame's coverage: gaps in training data produce systematic errors the same way frame undercoverage does. FSS training data may come from administrative records or prior survey waves, each with CIPSEA and quality implications.

Cited in: [nist_ai_rmf_100_1](#) , [advancing_american_ai_act_2022](#) , [americas_ai_action_plan_2025](#)

inference AI CORE ⚠ FALSE FRIEND

also: model inference

In AI, 'inference' means running a trained model to get an output for a new input — the production step, after training. It does NOT mean drawing conclusions about a population.

DEFINITION

The stage of ML in which a model is applied to a task. For example, a classifier model produces the classification of a test sample.

— NISTIR 8269 (draft), via NIST AI 100-3 (AIRC glossary) U.S. federal

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

Statistical inference quantifies uncertainty about a population parameter from a sample. ML inference is the forward pass of a fitted model on new inputs and carries no population-level uncertainty claim by itself.

✓ GROUNDED IN THE FSS CORPUS

A hard FALSE FRIEND. To a statistician, 'inference' means reasoning from a sample to a population (confidence intervals, hypothesis tests). To an ML engineer, 'inference' is just 'the model made a prediction.' Same word, different concept — confirm which sense is meant in any AI procurement or methods discussion.

Cited in: [fcs_m_20_04](#) , [m_24_10](#) , [m_25_21](#)

model drift AI CORE

also: drift, concept drift

When a deployed model's accuracy decays over time because the world it sees no longer matches the world it was trained on (populations shift, definitions change, behavior changes). It degrades silently unless you monitor for it.

DEFINITION

[concept drift] an online supervised learning scenario when the relation between the input data and the target variable changes over time.

— Gama et al. (concept drift), via NIST AI 100-3 (AIRC glossary) academic

Also defined: Systems that classify or predict a concept (e.g., credit ratings or computer intrusion monitors) over time can suffer performance loss when the concept they are tracking changes. — Raynor, via NIST AI 100-3 (AIRC glossary) academic

Also defined: Use of a system outside the planned domain of application, and a common cause of performance gaps between laboratory settings and the real world. — NIST SP 1270, via NIST AI 100-3 (AIRC glossary) U.S. federal

CROSSWALK → **temporal nonsampling error / frame obsolescence** (analog_of): Both are degradation of a fixed instrument as the target population moves. Disanalogy: frame obsolescence is auditable against updated frames; model drift needs explicit performance monitoring to detect.

✓ GROUNDED IN THE FSS CORPUS

Closely analogous to a sampling frame going stale or mode effects shifting across survey cycles — the instrument slowly stops measuring what it used to.

Cited in: `m_24_18`, `nist_ai_rm_f_100_1`, `nist_ai_rm_f_playbook`

synthetic data AI CORE

also: *synthetic microdata*

Artificially generated records that mimic the statistical properties of real data without being any real person's actual record. Used for privacy-protective sharing, testing, and model training.

DEFINITION

A process in which seed data are used to create artificial data that have some of the statistical characteristics of the seed data.

— NIST SP 800-188 U.S. federal

Also defined: a process in which seed data is used to create artificial data that has some of the statistical characteristics as the seed data — NISTIR 8053 U.S. federal

✓ GROUNDED IN THE FSS CORPUS

An emerging disclosure-avoidance tool: synthetic microdata products let researchers work with statistically representative data while protecting Title 13 / CIPSEA confidentiality. Utility and disclosure risk must both be assessed.

Cited in: `acdeb_final_report_2022` , `icsp_private_datasets_2023` , `nist_genai_profile_600_1`

differential privacy

AI CORE

CONTESTED

also: DP, formal privacy

A mathematical guarantee that adding or removing any one person's data barely changes what is published, so an attacker cannot reliably tell whether you were in the dataset. The protection is tunable by a 'privacy-loss budget' (epsilon): more privacy means more noise and less accuracy.

DEFINITION

A rigorous mathematical definition of disclosure that considers the risk that an individual's confidential data may be learned as a result of a mathematical analysis based on that data being made publicly available.

— NIST SP 800-188 [U.S. federal](#) · **CONTESTED**

Also defined: A mathematical framework that quantifies privacy risk to individuals as a consequence of data collection and subsequent data release. — NIST SP 800-226 [U.S. federal](#)

CROSSWALK → **statistical disclosure limitation (formal)** (instance_of): DP is a formal, provable form of SDL with a quantified privacy-accuracy trade-off. Disanalogy: traditional SDL (cell suppression, swapping) gives heuristic protection with no formal bound; DP gives a provable bound at an explicit, debated accuracy cost.

✓ GROUNDED IN THE FSS CORPUS

The Census Bureau's Disclosure Avoidance System applied formal differential privacy to 2020 Census products — the first nationwide federal use. NIST publishes two official definitions (SP 800-188 and SP 800-226); both are kept here. Politically contested inside Commerce (see DAO-216-26).

Cited in: [eo_14110](#) , [acdeb_final_report_2022](#) , [m_25_05](#)

de-identification

AI CORE

also: de-identify

Stripping or masking the parts of a record that point to a specific person, so the data can be used without exposing identities. Note: de-identified is not the same as re-identification-proof — combinations of remaining fields can still single someone out.

DEFINITION

General term for any process of removing the association between a set of identifying data and the data subject.

— NIST SP 800-188 (De-Identifying Government Datasets) [U.S. federal](#)

✓ GROUNDED IN THE FSS CORPUS

The umbrella practice under which suppression, swapping, noise injection, and differential privacy all sit. CIPSEA's 'identifiable form' standard (44 U.S.C. § 3561) is the legal target de-identification must defeat.

Cited in: [cipsea_2018](#) , [m_19_23](#) , [pra_1995_uslm](#)

re-identification AI CORE

also: re-identify

The reverse of de-identification: using leftover fields, outside data, or clever attacks to figure out who a 'de-identified' record actually is. FSS law defines the line in terms of whether identity can be 'reasonably inferred' by direct OR indirect means.

DEFINITION

"identifiable form" means any representation of information that permits the identity of the respondent to whom the information applies to be reasonably inferred by either direct or indirect means.

— CIPSEA 2018 §3561(7) — 'identifiable form' (the re-identification standard) U.S. federal

✓ GROUNDED IN THE FSS CORPUS

The legal harm CIPSEA confidentiality protections exist to prevent; the 'reasonably inferred by direct or indirect means' test is what disclosure-avoidance methods are measured against.

Cited in: sap_policy_m_23_04 , wp_22 , acdeb_final_report_2022

AI Governance & Risk

11 terms

high-impact AI

AI GOVERNANCE & RISK

△ FALSE FRIEND

also: high-impact AI use case

The federal government's label for the AI uses that can actually hurt people or violate rights, which therefore get the strictest oversight. Under the current memo (M-25-21) this single 'high-impact' category replaced the older split of 'safety-impacting' vs 'rights-impacting' (M-24-10).

DEFINITION

AI with an output that serves as a principal basis for decisions or actions that have a legal, material, binding, or significant effect on rights or safety.

— OMB M-25-21 (Apr. 3, 2025) [U.S. federal](#)

△ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

FALSE FRIEND: NIST's cybersecurity 'high-impact system' (FIPS 199) is about confidentiality/integrity/availability impact, NOT the OMB AI rights/safety sense. Don't conflate them.

✓ GROUNDED IN THE FSS CORPUS

Determines which agency AI tools trigger minimum risk-management practices (pre-deployment testing, impact assessment, ongoing monitoring). Most core statistical-production AI is not high-impact, but tools informing individual determinations can be.

Cited in: [m_25_22](#) , [m_25_21](#) , [eo_14110](#)

AI use case inventory AI GOVERNANCE & RISK

also: use case inventory

A yearly, public list every federal agency must publish of where it uses AI, flagging the high-impact uses and how it manages their risks. It is the main transparency mechanism for federal AI.

DEFINITION

An annual, public inventory of an agency's AI use cases, identifying high-impact use cases and describing associated risks and mitigations, required of federal agencies.

— OMB M-25-21 (Apr. 3, 2025) U.S. federal

✓ GROUNDED IN THE FSS CORPUS

Statistical agencies must inventory production AI tools — imputation models, automated coding, disclosure-avoidance systems — beginning with the 2024 cycle.

Cited in: [americas_ai_action_plan_2025](#) , [m_24_10](#) , [m_24_18](#)

Chief AI Officer AI GOVERNANCE & RISK

also: CAIO

The senior official each agency must name to own its AI strategy, governance, and risk. In statistical agencies this role overlaps with the Chief Data Officer and Statistical Official.

DEFINITION

A senior agency official designated under OMB AI guidance to coordinate and oversee the agency's use of AI, including its risk management and the AI use case inventory.

— OMB M-25-21 (Apr. 3, 2025); originally OMB M-24-10 U.S. federal

✓ GROUNDED IN THE FSS CORPUS

The CAIO's remit intersects the CDO and Statistical Official; in an FSS agency, AI-assisted collection and dissemination create shared responsibility across these roles.

Cited in: [m_24_10](#) , [m_25_21](#)

AI impact assessment AI GOVERNANCE & RISK

also: *AI impact analysis*

A required write-up, before you turn an AI tool on (and repeated periodically), of what it's for, who it could harm, and how you'll manage that. The AI equivalent of a methods-and-limitations review.

DEFINITION

A structured, pre-deployment and ongoing evaluation of an AI use case's intended purpose, expected benefit, and potential risks, required as a minimum practice for high-impact (formerly safety-/rights-impacting) AI.

— OMB M-25-21 (Apr. 3, 2025); cf. OMB M-24-10 §5 U.S. federal

CROSSWALK → **SPD No. 2 quality review / methodology review** (analog_of): Both are structured pre-release reviews of accuracy, fitness, and risk. Disanalogy: an AI impact assessment centers rights/safety harms and bias, which a classical quality review does not foreground.

✓ GROUNDED IN THE FSS CORPUS

Principal statistical agencies must complete these for qualifying tools.

Cited in: [m_25_21](#) , [m_25_22](#) , [nist_ai_rmf_100_1](#)

algorithmic bias AI GOVERNANCE & RISK

also: *AI bias*

When an AI system is systematically wrong in a way that disadvantages a particular group — often because the training data or design baked in an existing skew. It is not random error; it is patterned and tends to track demographic lines.

DEFINITION

Systematic and repeatable errors in an AI system that create unfair outcomes, such as privileging one group over another; arising from data, modeling choices, or human/societal factors (NIST distinguishes systemic, statistical/computational, and human-cognitive bias).

— NIST SP 1270 (Towards a Standard for Identifying and Managing Bias in AI) U.S. federal

CROSSWALK → **nonresponse bias / differential measurement error** (analog_of): Both are systematic (not random) distortions that hit specific subgroups. Disanalogy: nonresponse bias has an established estimation/adjustment theory; algorithmic bias often lacks a measurable true value to adjust toward.

✓ GROUNDED IN THE FSS CORPUS

Central risk when AI is used for demographic estimation, record linkage, or automated classification; can deepen undercoverage of underrepresented groups.

Cited in: [nist_ai_rmf_100_1](#) , [nist_ai_rmf_playbook](#)

AI red-teaming AI GOVERNANCE & RISK

also: red-teaming, red team

Deliberately attacking your own AI system — feeding it adversarial or edge-case inputs — to find where it fails, produces biased or harmful output, or can be manipulated, before real-world use. AI red-teaming adapts the cybersecurity practice to model behavior, not just security.

DEFINITION

A group of people authorized and organized to emulate a potential adversary's attack or exploitation capabilities against an enterprise's security posture. The Red Team's objective is to improve enterprise cybersecurity by demonstrating the impacts of successful attacks and by demonstrating what works for the defenders (i.e., the Blue Team) in an operational environment. Also known as Cyber Red Team.

— CNSSI 4009-2015 (cybersecurity sense) U.S. federal

CROSSWALK → **cognitive testing / respondent debriefing / reinterview** (analog_of): Both adversarially probe an instrument for failure modes before production. Disanalogy: cognitive testing studies human comprehension of questions; AI red-teaming studies model behavior under adversarial inputs.

✓ GROUNDED IN THE FSS CORPUS

Can stress-test FSS AI tools for demographic bias in classification, failure on edge-case data, or manipulation of public-facing NLP systems.

Cited in: eo_14110 , m_24_10 , m_24_18

explainability

AI GOVERNANCE & RISK

also: interpretability, explainable AI

Whether you can say, in terms a human can follow, WHY the AI produced a given output. A model can be accurate but unexplainable (a 'black box'), which is a problem when you must defend a methodology publicly.

DEFINITION

A property of AI systems concerning the degree to which the reasons for an output can be represented in terms meaningful to a given audience; one of the trustworthy-AI characteristics (explainable and interpretable).

— NIST AI 100-1 (AI Risk Management Framework 1.0) **U.S. federal**

CROSSWALK → **methodology documentation / transparency (SPD No. 2)** (analog_of): Both make a result auditable and reproducible to outsiders. Disanalogy: methodology docs describe a designed procedure; explainability must reconstruct the behavior of a learned model that has no hand-written procedure.

✓ GROUNDED IN THE FSS CORPUS

FSS uses of AI for record linkage, imputation, or small-area estimation must be explainable to data users and oversight bodies, consistent with SPD No. 2 transparency expectations.

Cited in: [americas_ai_action_plan_2025](#) , [m_25_22](#) , [nist_ai_rmf_100_1](#)

trustworthy AI

AI GOVERNANCE & RISK

⚠ FALSE FRIEND

also: responsible AI

NIST's checklist of the seven properties an AI system needs to be worthy of trust. It is the evaluative baseline federal agencies use to judge whether an AI tool is acceptable.

DEFINITION

AI that is valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed.

— NIST AI 100-1 (AI Risk Management Framework 1.0) **U.S. federal**

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

FALSE FRIEND: NIST's cybersecurity 'trustworthiness' is about security/reliability of an information system; 'trustworthy AI' is the seven-characteristic AI-RMF construct. Different frameworks.

✓ GROUNDED IN THE FSS CORPUS

The seven NIST characteristics form the assessment baseline for AI tools deployed by principal statistical agencies.

Cited in: [eo_14110](#) , [m_24_10](#) , [m_25_21](#)

confabulation

AI GOVERNANCE & RISK

⚠ FALSE FRIEND

also: fabrication

When a generative model states something false as if it were true — invented citations, wrong numbers, made-up facts — fluently and without warning. NIST prefers 'confabulation' over 'hallucination' because the model isn't perceiving anything; it is filling a gap with a plausible-sounding guess.

DEFINITION

The production of confidently stated but erroneous or false content (also called 'hallucination' or 'fabrication') by a generative-AI system; identified by NIST as a pervasive generative-AI risk.

— NIST AI 600-1 (Generative AI Profile) [U.S. federal](#)

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

Both are content errors — a reported value departs from the truth. Disanalogy: measurement error is bounded around a true value and modeled as a distribution; confabulation invents referents that do not exist, so it has no true value to be bounded around.

✓ GROUNDED IN THE FSS CORPUS

Critical risk when LLMs draft statistical reports, generate metadata, or summarize findings; mandates human verification before any figure or claim is relied on. FALSE FRIEND: an FSS reader may hear 'confabulation/hallucination' as harmless creativity; it is a specific risk of confident fabrication, and it is NOT measurement error (it can invent referents).

Cited in: [m_24_18](#) , [webb_nist_fcs_m_crosswalk](#) , [m_19_23](#)

information integrity

AI GOVERNANCE & RISK

The danger that AI-made content pollutes the information environment with convincing but false or manipulative material. For a statistical agency, the worry is AI-drafted public text that misstates the data.

DEFINITION

The risk that AI-generated content undermines accurate, truthful, and reliable information — including disinformation, propaganda, or manipulative content; generative-AI risk category in NIST AI 600-1.

— NIST AI 600-1 (Generative AI Profile) [U.S. federal](#)

✓ GROUNDED IN THE FSS CORPUS

Applies when agencies use generative AI for press releases, data-product narratives, or public-facing summaries — official statistics carry an authority that misinformation can borrow.

Cited in: [nist_genai_profile_600_1](#)

automated decision system AI GOVERNANCE & RISK

also: ADS, automated decision-making

A computer process that makes a consequential decision about a person — or substantially drives a human's decision — with little or no meaningful human judgment (e.g. auto-approving or flagging a case). The concern is consequential decisions handed to software without meaningful human review.

DEFINITION

a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.

— 15 U.S.C. § 9401(3) (National AI Initiative Act of 2020) U.S. federal

Also defined: OMB M-24-10 uses 'automated decision-making systems' as an operative term — e.g. in its definition of automation bias: 'the propensity for humans to inordinately favor suggestions from automated decision-making systems and to ignore or fail to seek out contradictory information.' — OMB M-24-10 (federal AI-governance usage)

U.S. federal

Also defined: any technology that either assists or replaces the judgement of human decision-makers. — Treasury Board of Canada Secretariat, Directive on Automated Decision-Making (2021) intl. government

✓ GROUNDED IN THE FSS CORPUS

No single U.S.-federal statutory definition of 'automated decision system' exists. The U.S.-federal anchors are the AI statutory definition (15 U.S.C. § 9401(3) — cited directly, NOT via the revoked EO 14110 that quotes it) and OMB AI-governance usage (M-24-10); the Canadian Treasury Board directive is the cleanest concise operational definition. Relevant in the FSS where data products feed individual-level determinations; pair with human-in-the-loop requirements.

Cited in: [m_24_10](#)

disclosure avoidance

FSS BRIDGE

CONTESTED

also: disclosure limitation

The whole set of methods agencies apply before releasing statistics so that no individual respondent can be identified — suppression, rounding, swapping, noise injection, differential privacy. Required by the confidentiality statutes.

DEFINITION

The set of methods to reduce the risk of disclosing information on individuals, businesses or other organizations. Such methods are only related to the dissemination step and are usually based on restricting the amount of or modifying the data released.

— NIST SP 800-188 (statistical disclosure control) **U.S. federal** · **CONTESTED**

CROSSWALK → **differential privacy** / **cell suppression** / **swapping** (umbrella_of): Disclosure avoidance is the goal; DP, suppression, swapping, and noise injection are alternative methods that achieve it with different utility/privacy trade-offs.

✓ GROUNDED IN THE FSS CORPUS

Required by Title 13, CIPSEA, and agency statutes. The Census Bureau's Disclosure Avoidance System (DAS) implemented differential privacy at scale for the 2020 Census. The choice of method is politically contested inside Commerce (see DAO-216-26).

Cited in: `acdeb_final_report_2022` , `cipsea_2002_implementation_guidance` , `fcs_m_20_04`

statistical disclosure limitation FSS BRIDGE

also: *SDL, statistical disclosure control*

The technical name for the family of methods that reduce the risk that a published statistic reveals information about a specific individual or business. In the FSS this is the methodological core of disclosure avoidance.

DEFINITION

The set of methods to reduce the risk of disclosing information on individuals, businesses or other organizations. Such methods are only related to the dissemination step and are usually based on restricting the amount of or modifying the data released.

— NIST SP 800-188 U.S. federal

CROSSWALK → **disclosure avoidance** (synonym_near): Used nearly interchangeably; 'disclosure avoidance' is the Census Bureau's operational term, 'statistical disclosure limitation/control' the methodological literature's term.

✓ GROUNDED IN THE FSS CORPUS

The canonical FSS reference is FCSM Statistical Policy Working Paper 22, 'Report on Statistical Disclosure Limitation Methodology' (in this corpus as wp_22). Differential privacy is a formal member of this family.

Cited in: [acdeb_final_report_2022](#) , [m_25_05](#) , [wp_22](#)

measurement error FSS BRIDGE

The gap between the answer recorded and the true value, caused by the respondent, interviewer, question wording, mode, or records. One of the components of nonsampling error.

DEFINITION

Measurement error is the difference between the true value of the measurement and the value obtained during the measurement process.

— U.S. Census Bureau Statistical Quality Standards — Glossary U.S. federal

CROSSWALK → **confabulation (generative AI)** (analog_of): Both are content errors relative to a true value. Disanalogy: measurement error is bounded by a true value; confabulation can invent non-existent referents.

✓ GROUNDED IN THE FSS CORPUS

A core FSS quality concept (U.S. Census Bureau Statistical Quality Standards; the D3 standard in this corpus is about producing measures of it). Its AI relevance: NLP tools parsing open-ended responses can introduce systematic measurement error that varies by respondent group, which is the bridge to AI confabulation/bias.

Cited in: [fcs_m_20_04](#) , [wp_31](#) , [wp_47](#)

nonsampling error FSS BRIDGE

Every source of error in a statistic that is NOT from random sampling — coverage gaps, nonresponse, measurement error, processing mistakes. Often the dominant error in large surveys, and harder to quantify than sampling error.

DEFINITION

Nonsampling errors are survey errors caused by factors other than sampling (e.g., nonsampling errors include errors in coverage, response errors, nonresponse errors, faulty questionnaires, interviewer recording errors, and processing errors).

— U.S. Census Bureau Statistical Quality Standards — Glossary U.S. federal

CROSSWALK → **model error / bias + drift (AI)** (analog_of): Nonsampling error is the FSS's mature taxonomy of non-random error; AI bias, drift, and confabulation map onto its branches (coverage, temporal, measurement).

✓ GROUNDED IN THE FSS CORPUS

The FSS error framework (Census Statistical Quality Standards; the D3 standard in this corpus) that AI systems must be evaluated against — AI can both introduce and mitigate nonsampling error, so it must be judged on all these dimensions, not just predictive accuracy.

Cited in: [wp_31](#) , [census_sqs_d3](#) , [census_sqs_f1](#)

data asset FSS BRIDGE ⚠ FALSE FRIEND

In FSS law, simply a defined collection of data that can be grouped and managed as a unit. It is a governance/inventory term, not a technical or security term.

DEFINITION

the term 'data asset' means a collection of data elements or data sets that may be grouped together.

— CIPSEA 2018 / Evidence Act §3502 (44 U.S.C. § 3502) U.S. federal

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

FALSE FRIEND: NIST/CNSSI 4009 treats a 'data asset' as something to be secured (a cyber asset). The FSS statutory sense is a governed collection of data for inventory/access purposes. The obligations differ entirely.

✓ GROUNDED IN THE FSS CORPUS

Carried by CIPSEA/Evidence Act §3502 and referenced across OMB memos (M-24-10, M-25-21, M-23-04); 80+ obligations in this corpus attach to 'data asset' (open-data, inventory, and Standard Application Process access duties).

Cited in: [cipsea_2018](#) , [cipsea_2018_uslm](#) , [evidence_act_2018](#)

confidentiality

FSS BRIDGE

⚠ FALSE FRIEND

In FSS law, a binding duty NOT to disclose protected information to anyone unauthorized — a legal obligation, not just a security setting. CIPSEA makes willful unauthorized disclosure a federal crime.

DEFINITION

The term 'confidentiality' means a quality or condition accorded to information as an obligation not to disclose that information to an unauthorized party.

— CIPSEA 2018 §3561(2) (44 U.S.C. § 3561) [U.S. federal](#)

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

FALSE FRIEND: in cybersecurity 'confidentiality' is one leg of the CIA triad (a technical access-control property). In FSS law it is a statutory non-disclosure obligation with criminal penalties. Related spirit, very different force.

✓ GROUNDED IN THE FSS CORPUS

The foundational FSS duty. Any AI use of statistical data inherits it: a model trained on confidential data cannot be deployed in a way that permits re-identification.

Cited in: [cipsea_2018](#) , [cipsea_2018_uslm](#) , [eo_14110](#)

statistical product

FSS BRIDGE

⚠ CONTESTED

also: statistical products

The published outputs of a statistical agency — the tables, estimates, and microdata it releases. FSS rules require they be produced with quality, released equitably, and documented transparently.

DEFINITION

Statistical products are the data and information (tables, estimates, microdata, analyses) disseminated by a statistical agency; the Trust Regulation frames duties of equitable access to and transparency about statistical products.

— Fundamental Responsibilities of RSAUs — 'Trust Regulation' (5 CFR Part 1321) [U.S. federal](#) · [CONTESTED](#)

✓ GROUNDED IN THE FSS CORPUS

Governed by SPD No. 1/No. 4 and the 2024 Trust Regulation (5 CFR 1321); the disclosure-avoidance method applied to a statistical product is the live policy question in DAO-216-26.

Cited in: [trust_regulation_nprm](#) , [spd_1](#) , [spd_4](#)

noise injection

FSS BRIDGE

CONTESTED

also: *noise infusion*

A disclosure-avoidance technique that protects respondents by adding small random values to data before release, so no single published number pins down one person. Differential privacy is a formal, calibrated way of doing this.

DEFINITION

A de-identification technique that modifies a dataset by adding random values to the values of a selected attribute.

— NIST SP 800-188 **U.S. federal** · **CONTESTED**

CROSSWALK → **differential privacy** (method_family): Differential privacy is a formal, provably-calibrated form of noise injection; 'noise injection' also covers informal/heuristic noise addition without a formal privacy guarantee. M-25-05 footnote 89 names the two together.

✓ GROUNDED IN THE FSS CORPUS

NIST defines it as a de-identification technique (SP 800-188). In the FSS it is one method of disclosure avoidance, closely tied to differential privacy. **CONTESTED**: DAO-216-26 (DOC, June 2026) bans 'noise injection' while OMB M-25-05 permits the technique with documentation — the conflict the KG surfaced (see docs/2026-06-22_lab_notes_m25_05_s317_closed.md). Stated neutrally; not adjudicated here.

Cited in: `m_25_05` , `webb_nist_fcsn_crosswalk`

AI-ready data

FSS BRIDGE

★ FLAGSHIP

also: AI-readiness, AI-ready

Data is “AI-ready” when a machine — typically an LLM — can not just open it but actually understand it: the numbers arrive with enough context, documentation, provenance, and quality information that the model interprets them correctly and answers questions without mangling the statistics. Machine-readable (a clean CSV) is necessary but not sufficient; machine-understandable is the bar. Note this assumes a system can already get to the data — whether it can find and retrieve it in the first place is a separate question (see the addendum).

DEFINITION

Data is AI-ready when it is machine-understandable — not merely machine-readable: its context, provenance, methodology, and quality signals are preserved and programmatically queryable, sufficient for an AI system (particularly a large language model) to analyze and answer questions about the asset without loss of statistical integrity. Per the federal project definition, AI-readiness is “the extent to which a data asset is prepared for effective analysis and querying by AI systems, particularly LLMs,” spanning data quality, access, formatting, and metadata. Scope boundary: this definition assumes access to the data asset is already established — it addresses the readiness of data that a consuming system can already reach. (The discoverability/retrievability axis is treated separately in the forward-interpretation addendum, deliberately outside the grounded definition.)

— America’s DataHub RFS — Topic MLMU-25, “Measuring Large Language Model Understanding of Federal Statistical Data” (footnote 2); framed as an extension of FCSM 25-03 (AI-Ready Federal Statistical Data) and FCSM 20-04 (A Framework for Data Quality). **U.S. federal**

Scope boundary: Definition (Part A) covers the readiness of data a consuming system can already REACH; it assumes access is established. The discoverability/retrievability (access) axis is addressed only in the forward-interpretation addendum (Part B), explicitly ahead of policy.

Also defined: For the purposes of this project, “AI-readiness” refers to the extent to which a data asset is prepared for effective analysis and querying by AI systems, particularly LLMs. This includes data quality, access, formatting, and metadata. AI-ready data enable LLMs to generate meaningful insights from both structured (e.g., tabular) and unstructured (e.g., reports) formats by facilitating accurate interpretation and contextual understanding. — America’s DataHub RFS, Topic MLMU-25, footnote 2 (verbatim, verified against source 2026-06-23) **U.S. federal**

CROSSWALK → **Lawrence Data Readiness Levels (2017), Bands C/B/A — academic conceptual ancestor (arXiv:1705.02245)** (descends_from): Lawrence’s Data Readiness Levels are the conceptual ancestor: Band C = accessibility (can the data be accessed), Band B = faithfulness/validity (is it what it claims to be), Band A = data-in-context (ready for a specific, defined task). AI-readiness inherits the use-case-conditional logic — Lawrence shows readiness is relative to the intended use (this use-case-conditionality point is Lawrence’s, 2017; it is NOT attributed to Gartner). Disanalogy: DRL is a general-purpose data-maturity scale; this definition is scoped to federal statistical data and adds the “without loss of statistical integrity” constraint a general maturity scale does not carry. Cited as an external ancestor only — academic, not federal policy, so not ingested into the corpus.

✓ GROUNDED IN THE FSS CORPUS

Grounded in the corpus: the federal definition (America’s DataHub / NSDS RFS) plus the FCSM 25-03 AI-ready extension and the FCSM 20-04 data-quality framework. The corpus establishes that creating AI-ready data “builds on established open-data principles” and machine-readability (FCSM 25-03 §s34), that federal

statistics used in AI interactions must meet the Framework’s data-quality dimensions — accessibility, timeliness, accuracy (§s30) — and that metadata enrichment exposed through AI-friendly APIs improves how LLMs discover and interpret data (§s31). FCSM 20-04 §s184 anchors the data-quality dimensions that must be balanced and preserved.

Cited in: [bea_rfs_mlm25](#) , [fcs25_03](#) , [fcs25_03](#) , [fcs25_03](#) , [fcs20_04](#)

▶▶ **FORWARD INTERPRETATION — AHEAD OF DOCUMENTED POLICY (NOT AN ESTABLISHED STANDARD)** ·
dated 2026-06

Practitioner framing (NOT graph-grounded; ahead of the documented standard): the federal and Commerce/FCSM definitions above assume access to the data asset is already established. In practice, access and machine-retrievability must be deliberately configured — via emerging mechanisms such as llms.txt, the Model Context Protocol (MCP), and WebMCP, the last standardized only around 2026, i.e. AFTER the FCSM/Commerce literature was authored. Written policy lags the technology in this fast-moving field. A complete AI-data-readiness picture therefore treats discoverability and retrievability as a PRIOR axis the current standards do not yet cover: data a model cannot find or reach is not AI-ready in any operational sense, however machine-understandable it is once reached. This is the machine-as-first-class-user position, offered as a forward-looking, bounded thought-leadership seed — not a mandate, and deliberately separated from the grounded definition so the definition’s authority is not diluted by interpretation.

Builds on (does not supersede): FAIR (Findable, Accessible, Interoperable, Reusable); FCSM 20-04 (A Framework for Data Quality); FCSM 25-03 (AI-Ready Federal Statistical Data). FAIR — AI-readiness builds on / extends FAIR; it does not supersede it.

High Translation-Load

9 terms

hallucination

HIGH TRANSLATION-LOAD

⚠ FALSE FRIEND

The popular word for an AI confidently making something up. Prefer 'confabulation' in official writing — NIST does, because the model isn't perceiving anything, it's guessing.

DEFINITION

Colloquial term for confidently stated but false content generated by a model; NIST uses 'confabulation' (and 'fabrication') in preference to 'hallucination.'

— NIST AI 600-1 (Generative AI Profile) [U.S. federal](#)

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

See confabulation: error vs. a true value, except confabulation/hallucination can invent referents with no true value.

✓ GROUNDED IN THE FSS CORPUS

Same risk as confabulation; the term you'll hear from vendors and the press. FALSE FRIEND: not a perceptual glitch and not measurement error — a confident fabrication. See [confabulation].

Cited in: [m_24_18](#) , [nist_genai_profile_600_1](#) , [webb_nist_fcs_m_crosswalk](#)

bias

HIGH TRANSLATION-LOAD

⚠ FALSE FRIEND

A loaded word that means different things to different audiences. To a statistician, bias is a precise, often quantifiable concept: the expected difference between an estimator and the true parameter (and can be desirable, e.g., a bias-variance trade-off). In AI/ethics discourse, 'bias' usually means unfair, discriminatory outcomes.

DEFINITION

Bias is the difference between the expected value of an estimator and the actual population value.

— U.S. Census Bureau Statistical Quality Standards — Glossary (statistical sense) [U.S. federal](#)

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

Statistical bias is $E[\text{estimator}] - \text{true value}$, a neutral quantifiable quantity. AI/algorithmic bias is normative — patterned unfair outcomes for groups. Same word, different field, different valence.

✓ GROUNDED IN THE FSS CORPUS

A hard FALSE FRIEND — clarify which sense is meant. Statistical bias (estimator vs. parameter) is not the same as algorithmic/social bias (unfair group outcomes). NIST SP 1270 catalogs the AI senses; pull its wording for the AI definition.

Cited in: [webb_nist_fscm_crosswalk](#) , [americas_ai_action_plan_2025](#) , [eo_14179](#)

validity

HIGH TRANSLATION-LOAD

⚠ FALSE FRIEND

Whether the system actually measures the thing you think it measures (not just consistently — that's reliability). A model can be reliably wrong.

DEFINITION

Validity (with reliability) is a trustworthy-AI property: the degree to which an AI system measures or produces what it is intended to, confirmed by objective evidence.

— NIST AI 100-3 (The Language of Trustworthy AI) [U.S. federal \(beta\)](#)

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

Both ask 'are we measuring the intended construct.' Largely compatible across the survey-methodology and AI senses; the survey literature splits it further (content/construct/criterion validity).

✓ GROUNDED IN THE FSS CORPUS

Maps closely to FSS notions of accuracy/unbiasedness under SPD No. 2. NIST AI 100-3 is provisional (beta) — verify exact wording before formal reliance.

Cited in: [webb_nist_fscm_crosswalk](#) , [m_24_10](#) , [m_25_21](#)

reliability

HIGH TRANSLATION-LOAD

⚠ FALSE FRIEND

Whether the system gives the same answer under the same conditions — consistency.
Reliable but invalid means consistently measuring the wrong thing.

DEFINITION

The probability of performing a specified function without failure under given conditions for a specified period of time.

— NIST IR 8401 **U.S. federal**

⚠ FALSE FRIEND — DIFFERENT IN AI VS. THE FSS

Compatible: both mean consistency of results under repetition.

✓ GROUNDED IN THE FSS CORPUS

In FSS quality terms, reliability maps to reproducibility across processing cycles. Note NIST uses 'reliability' in both a systems-engineering sense and a trustworthy-AI sense.

Cited in: `americas_ai_action_plan_2025` , `fcs_m_20_04` , `privacy_act_1974_uslm`

model card

HIGH TRANSLATION-LOAD

also: system card

A standardized 'nutrition label' for an AI model: what it's for, how well it works, on whom it was tested, and where it fails. Agencies should demand one before adopting a model.

DEFINITION

A short documentation artifact accompanying a trained model that reports its intended use, performance across conditions, evaluation data, limitations, and ethical considerations.

— Mitchell et al., 'Model Cards for Model Reporting' (FAT* 2019) **academic**

CROSSWALK → **survey methodology documentation / technical documentation (SPD No. 2)** (analog_of): Both document methodology, performance, and appropriate-use conditions. Disanalogy: methodology docs describe a designed procedure; a model card reports empirical behavior of a learned model.

📌 PRACTITIONER NOTE — NOT CORPUS-SOURCED

Practitioner framing (NOT graph-grounded): the FSS should require a model card (and system card) as acquisition documentation for any adopted AI model — the AI analog of survey methodology documentation. The corpus references 'system cards' only in passing (M-24-10) and a 'GPT-4 System Card' citation; it neither defines nor mandates model cards, so this is a recommended practice, not a corpus obligation.

provenance HIGH TRANSLATION-LOAD

also: data lineage, data provenance

The documented history of where data (or a model's training data) came from and how it was transformed — the chain of custody. Without it, you can't audit or reproduce a result.

DEFINITION

In the context of computers and law enforcement use, it is an equivalent term to chain of custody. It involves the method of generation, transmission and storage of information that may be used to trace the origin of a piece of information processed by community resources.

— CNSSI 4009-2015 U.S. federal

CROSSWALK → **metadata / data documentation** (component_of): Provenance is the origin-and-transformation portion of the broader metadata an FSS product must carry.

✓ GROUNDED IN THE FSS CORPUS

In the FSS, provenance/metadata is required for data products (CIPSEA defines 'metadata' to include provenance, 44 U.S.C. § 3561(19)); for AI it extends to model and training-data lineage under data-governance requirements.

Cited in: [cipsea_2018](#) , [eo_14110](#) , [evidence_act_2018](#)

human-in-the-loop HIGH TRANSLATION-LOAD

also: human oversight, human-on-the-loop

A design where a person reviews, approves, or can override the AI before its output takes effect — as opposed to fully automated action. 'On-the-loop' is lighter (monitoring) than 'in-the-loop' (approval). The point is meaningful human judgment, not a rubber stamp.

DEFINITION

An AI system that requires human interaction.

— DoD Modeling & Simulation Glossary, via NIST AI 100-3 (AIRC glossary) U.S. federal

CROSSWALK → **analyst review / adjudication in survey processing** (analog_of): Both insert human judgment before an automated output is accepted. Disanalogy: automation bias makes the human review weaker than it looks as volume scales.

✓ GROUNDED IN THE FSS CORPUS

The control that keeps AI-assisted coding, imputation, or edit-flagging from acting unreviewed; the risk is automation bias — humans rubber-stamping AI when review volume exceeds what they can meaningfully audit. OMB M-25-21 and NIST AI 600-1 (Human-AI Configuration) are the references — pull exact wording.

Cited in: [webb_nist_fesm_crosswalk](#) , [m_24_10](#) , [m_25_21](#)

fitness for use HIGH TRANSLATION-LOAD

also: *fitness for purpose*

The principle that 'quality' isn't absolute — data are good enough only relative to a specific purpose. The same dataset can be fit for one use and unfit for another. This is the lens for judging whether an AI tool or its data is 'good enough' for a given task.

DEFINITION

Quality is defined as fitness for use: the degree to which data are suitable for the intended use(s), assessed across multiple quality dimensions (relevance, accuracy, timeliness, accessibility, interpretability, coherence).

— FCSM, A Framework for Data Quality (FCSM 20-04) U.S. federal

CROSSWALK → **fitness for use (KG quality framing)** (same_concept): Identical principle the KG quality battery uses — quality is purpose-relative, so the user selects the dimensions that matter.

✓ GROUNDED IN THE FSS CORPUS

The FSS's governing definition of data quality (FCSM Data Quality Framework; this corpus holds FCSM 20-04 and 23-02). It is also the right frame for evaluating AI outputs: fit for which decision?

Cited in: [fcs_m_20_04](#) , [acdeb_final_report_2022](#) , [m_14_06](#)

prompt HIGH TRANSLATION-LOAD

also: *prompting, prompt engineering*

The text you type to an AI tool to tell it what you want; 'prompting' is the process of writing and providing such inputs to guide a generative model toward specific outputs. The same model gives very different answers to differently worded prompts — useful, and a reproducibility hazard.

DEFINITION

[prompts are] natural language requests that elicit accurate, high quality responses from a language model.

— Google Gemini API documentation, 'Prompt design strategies' industry

🔪 PRACTITIONER NOTE — NOT CORPUS-SOURCED

Practitioner framing (NOT graph-grounded): a prompt is the actual interface to a generative tool, so for an FSS AI-readiness review the concern is operational — an undocumented prompt makes an AI-assisted result irreproducible, and a prompt bearing confidential text can leak it to a third-party model. Not an FSS policy term; the corpus word-matches for 'prompt' are the English adjective, not this concept.